



Scanning

After footprinting scanning is the second phase of information gathering that hackers use to size up a network. Scanning is where they dive deeper into the system to look for valuable data and services in a specific IP address range.

In scanning part identify the Live system, open ports, Services, OS, Network Scan, Vulnerability scan.

`/usr/share/nmap/scripts`

Lua Language

For Local/Network network scan :

`#arp-scan -l`

`#netdiscover -r <ip range>`

`#nmap -sn <ip range>`

Port Status Types

With nmap you would see one of four port status types:

Open - It means that the port is accessible and an application is listening on it.

Closed - It means that the port is inaccessible and no application is listening on it.

Filtered - It means that nmap is not able to figure out if the port is open or closed, as the packets

are being filtered, which probably means that the machine is behind a firewall.

Unfiltered — It means that the ports are accessible by nmap but it is not possible to figure out if

they are open or closed.



Before understanding how port scanning works, we need to understand how the TCP three-way

handshake works.

The transmission control protocol (TCP) was made for reliable communication. It is used for a

wide variety of protocols on the Internet and contributes toward reliable communication with the

help of the three-way handshake.

SYN

SYN/ACK

ACK

First host sends a SYN packet to the second host.

Second host responds with a SYN/ACK packet, it indicates that the packet was received.

First host completes the connection by sending an acknowledgment packet.

TCP Flags

SYN - Initiates a connection.

ACK - Acknowledges that the packet was received.

RST - Resets the connections between two hosts.

FIN - Finishes the connection.

PSH - Sends all data immediately.

URG - Data contained in the packet should be processed immediately.

Nmap commands :

#nmap <ip address> - For Normal scan

#nmap <ip address> -v - Normal scan with verbosity

#nmap <ip address> -Pn - Use this if target system is blocking your request



```
#nmap <ip address> 21,22,25,80 -v -p - For scanning specific ports
```

```
#nmap <ip address> -p 0-65535 - For scanning all ports
```

```
#nmap <ip address> -sV - For services and their version
```

```
#nmap <ip address> -A - For aggressive/all scan
```

Scanning your target network through NMAP Timing Technique

The timing technique is one of the best techniques to evade firewalls/IDS. The idea behind this

technique is to send the packets gradually, so they do not end up being detected by firewalls/IDS.

In nmap we can launch a timing scan by specifying the T command followed by a number ranging

from 0 to 5. Increasing the values from T0 to T5 would increase the speed of the scan.

- ■ -T0—Paranoid
- ■ -T1—Sneaky
- ■ -T2—Polite
- ■ -T3—Normal
- ■ -T4—Aggressive
- ■ -T5—Insane

open a terminal in kali linux follow the steps to perform scanning -

```
# Nmap -help (to check the other options regarding Nmap)
```

```
# Nmap -A -T4 -r 192.168.0.1/24
```

```
# Nmap -A -T4 -r 192.168.0.1/24 -D Local IP (attacker IP) = -D to hide your IP to be detected by the IDS of the victim's PC
```

```
# Nmap -A -T4 -r 192.168.0.1/24 -oG <name of the file> = -oG <name of the file> after scanning to create a file for storing the data output.
```

```
# Nmap -A -T4 -r 192.168.0.1/24 > <name of the file> = > to specify the file.
```

Some Other Tools for scanning

Superscan3.0 - windows base tools to gather information about the target on the network to scan.

AngryIPscanner - Both available for Linux and windows

Nmap - tool is used to identify the target server like OS, open ports, service, vulnerability of the individual server.

Zenmap - Graphical version of Nmap